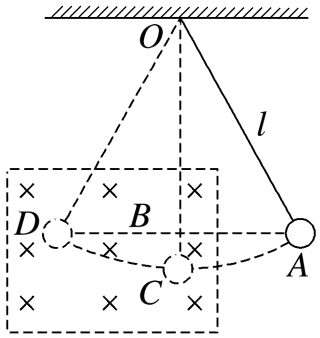
20.2.1 Exercises - Lorentz Force on a moving charge 1

1. A charged insulated ball of mass m (can be regarded as a point mass) is suspended at point O by an insulated string of length L, with a uniform magnetic field below the suspension point O. The ball is then released from rest at point A after being pulled away from its equilibrium position.



a. Compare and explain the speed at C when the ball travels from A to C and from D to C.

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b. Compare and explain the tension in the string at C when the ball travels from A to C and from D to C.

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c. Compare and explain the acceleration at C when the ball travels from A to C and from D to C.

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d. State and explain whether the mechanical energy of small ball due to the decreases, increases or stays constant as it travels from A to D.

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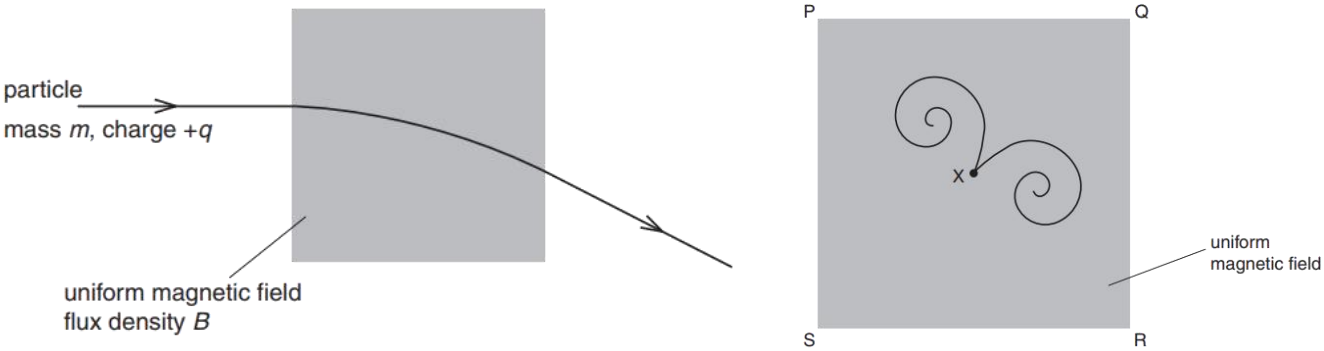
2.

a. A charged particle of mass m and charge +q is travelling with velocity v in a vacuum. It enters a region of uniform magnetic field of flux density B as shown in Fig.



The magnetic field is normal to the direction of motion of the particle. The path of the particle in the field is the arc of a circle of radius r.

i. Explain why the path of the particle in the field is the arc of a circle.

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ii. Show that the radius r is given by the expression[1]

$$r = \frac{mv}{QB}$$

b. A uniform magnetic field is produced in the region PQRS, as shown in Fig. (right)

The magnetic field is normal to the page.

At point X, a gamma-ray photon interaction causes two particles to be formed. The paths of these particles are shown in Fig. (right)

i. Suggest, with a reason, why each of the paths is a spiral, rather than the arc of a circle.

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ii. State and explain what can be deduced from the paths about

a) the charges on the two particles,

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.....[2]

b) the initial speeds of the two particles.

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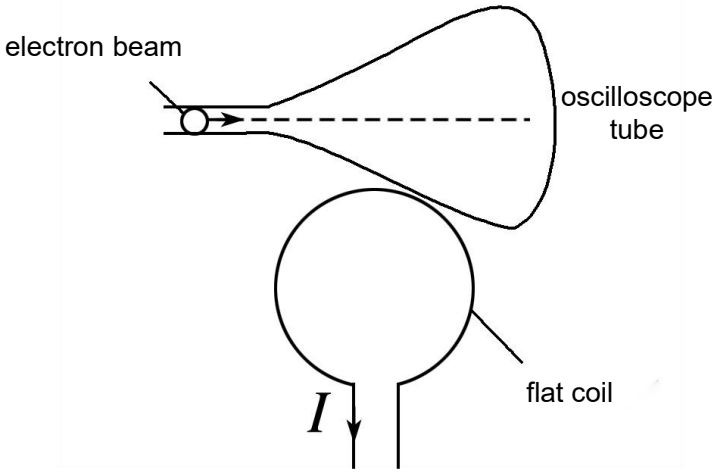
3. As shown in the fig., The electron beam emitted by the electron gun enters the oscilloscope tube. Directly below the oscilloscope tube, there is a vertically placed current-carrying flat coil.



a. Draw the magnetic field due to the current in the flat coil.

b. Draw the path of the electron beam in the oscilloscope tube.